

Case study: HealthSense

Liam, a software team lead at the health tech start-up VitalWell, is leading the development of HealthSense, an app that predicts potential illness – like the cold, the flu or COVID-19 – 1-3 days in advance by analyzing data from smartwatches and fitness trackers. HealthSense gathers biometric information such as heart rate variability, body temperature, and sleep patterns to build a personalized health profile, adjusting and improving over time.

The main goal of HealthSense is to provide users with early illness alerts, enabling proactive measures to prevent potentially spreading illnesses to others. For example, when the app identifies a heightened risk, users may choose to work from home, reducing the risk of exposure to coworkers. HealthSense uses a recurrent neural network (RNN) model with three layers of 128 GRU cells, allowing it to track and respond to subtle pattern changes in users' health data over several weeks.

To achieve these insights, HealthSense requires continuous data collection, storing detailed user health data to refine and personalize its time-series model. The app stores data on the user's device and, to enhance predictions, regularly syncs it with VitalWell's servers, where it is stored anonymously and periodically incorporated into a larger training dataset for model improvements.

The dataset for training HealthSense comprises health data from approximately 50,000 users, including a balanced demographic representation of age, gender, and pre-existing health conditions. This includes individuals aged 18 to 75, with approximately 15 % reporting chronic health issues such as asthma, diabetes, or heart disease.

In initial pilot testing, users found HealthSense useful, with a significant percentage logging into the app daily and taking precautions on days with higher illness predictions. Though the pilot feedback was largely positive, some users expressed reservations about the app's handling of data, particularly regarding continuous background data collection and data storage practices. Below is a summary of key performance metrics on the test set after 2 months of use (the numbers increase significantly as the model is personalized, starting around 0.75-0.8). The numbers are very similar for the train set.

Metric	Overall	Gender		Age		Chronic conditions?	
		Male	Female	<50	>50	Yes	No
Support	9,302	4,637	4,665	5,125	4,177	1,402	7,900
Accuracy	0.97	0.97	0.98	0.98	0.95	0.96	0.98
Precision	0.98	0.98	0.97	0.99	0.96	0.97	0.98
Recall	0.93	0.92	0.93	0.94	0.91	0.87	0.94

Note that *support* is the number of data points in the test set belonging to the indicated category. *Accuracy*, *recall* and *precision* are calculated with being sick as the positive class.

With a target launch set in the coming months, Liam needs to assess if HealthSense is ready for widespread release. You are to assist Liam in making his decision. Based on the information provided, **perform an ethical impact assessment and recommend how Liam should proceed**. Should HealthSense be deployed as is, or should Liam and his team take additional steps before launching – or abandon the project altogether?